

Luiz Sérgio da Silva Filho

SOFTWARE ENGINEER | FULL STACK DEVELOPER

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Open to hybrid, remote or on-site roles

SUMMARY

Information Technology graduate with hands-on experience in IT support, now focused on building full stack software that applies artificial intelligence in practical ways: structured document extraction, tool-calling agents, process automation and API integration. While running my own business, I independently build complete systems with databases, authentication, payments and real integrations, including a system running in production with paying customers and an AI agent that handles sales over WhatsApp. I apply engineering practices such as multi-tenant data isolation with Row-Level Security, automated integration tests and cost control on AI calls.

PROFESSIONAL EXPERIENCE

Pedagogical Coordinator

Feb 2026 – Present

Children's Programming and Robotics School · Barretos, SP, Brazil

- Built the pedagogical coordination from the ground up when the course opened: organized the classes, defined the curriculum by age group and assembled the teaching team, serving students from 7 to 18 years old.
- Age-progressive curriculum: younger classes work with robotics (Arduino, micro:bit) and block-based programming; teenage classes start with Python and move on to Git, HTML, CSS, JavaScript, Tailwind CSS, daisyUI, React, Node.js, MongoDB, JWT and Swagger as their full stack foundation.
- Teach Python to the teenage classes, covering programming logic, basic data structures and small hands-on projects.
- Faced with the limitations of the system the unit was using for pedagogical management and tracking, I ran the entire lifecycle of a replacement system on my own: gathered the requirements from my own coordination routine, designed the data model and the role-based permissions (director, coordinator, teacher), planned the delivery phases and validated every release against real school usage before putting it into production.
- The artificial intelligence layer built into the pedagogical workflow was designed with automatic fallback across multiple LLM providers for resilience, and with personal data anonymized before anything reaches the AI.

Owner and Partner

Jun 2020 – Present

Empório do Bebê · Barretos, SP, Brazil

- Responsible for running the business: point of sale and inventory control, online sales channel and digital catalog, and automation of administrative processes through spreadsheets.
- In parallel, independent development of a line of AI-powered software projects: systems with databases, authentication, payments and real integrations, including customer service automation, AI-driven data extraction and one system running in production with real customers.
- Multi-tenant data isolation with Row-Level Security in defense in depth, real integration tests, cost engineering on AI calls and active defense against prompt injection.

IT Support Analyst

Aug 2013 – Dec 2019

Sesi São Paulo · Limeira and Barretos, SP, Brazil

- IT support for staff, teachers and students: helpdesk ticket handling, administrative and educational systems, and support for classroom technology.
- Technical mentoring for educational robotics teams: support with programming and with planning the execution order of the robot missions to solve the proposed challenges, in competitions such as FLL (First Lego League) and OBR (Brazilian Robotics Olympiad).

INDEPENDENT TECHNICAL PROJECTS

Built independently, alongside my work as a partner at Empório do Bebê.

- Identity document verification system: extracts data from photographed documents using computer vision and checks the information for authenticity (such as national ID checksums and date validity) server side, without trusting the AI reading alone.
- AI sales agent that serves customers over WhatsApp: looks up stock and pricing in real time and closes orders through real tool calls against the database, with a safety guard that never accepts a price suggested by the AI without confirming it in the system.
- Multi-company SaaS platform: every client uses the same system with data fully isolated from one another (multi-tenant architecture), protected by Row-Level Security at the database and validated by an automated integration test suite.
- Inventory liquidation system in production: organizes products into lots and cards and runs auditable draws with cryptographically generated random numbers, keeping the full financial history in an append-only ledger. Already serving real customers.
- Recurring practice across all AI projects: I use multiple LLM providers (Claude and OpenAI) interchangeably, with per-call cost control, prompt caching to reduce latency and active defense against prompt injection.

EDUCATION

Postgraduate Degree in Higher Education Teaching 2015

Faculdade XV de Agosto

Bachelor's Degree in Information Technology 2008 – 2011

Faculdade Barretos

TECHNICAL SKILLS

Frontend & languages: TypeScript, Python, React.js, Next.js

Backend & APIs: Node.js, NestJS, REST API, Webhooks

Data & architecture: PostgreSQL, Prisma ORM, Drizzle ORM, Row-Level Security, Multi-tenant Architecture, Redis

Artificial intelligence: Anthropic Claude API, OpenAI API, LLM Integration, Prompt Engineering, AI-Assisted Development

Infrastructure & engineering practices: Docker, CI/CD, Git, BullMQ, Automated software testing, Stripe API, Process automation, n8n